



PDQ - WEEKLY FINDINGS

Changing the tech, not the thresholds

What the system has that the probe didn't, the probe rebuilt from those pieces, and what it does to the wrong answers. Plus: the edits test, and two holes fixed.

Nada. For the meeting on 4 October 2026.

The week in short

Wrong answers, pod's rules	7
Wrong answers, new probe + rules	1
Correct answers kept	48 of 52
Mirrored clips identified	0 of 8 before, 8 of 8 now
Letterboxed clips identified	1 of 8 before, 6 of 8 now

- Last week's advice was to change the tech rather than keep bending the rule thresholds.

So I compared the three parts of the engine that can answer, found what the careful part has that the deciding parts don't, and rebuilt the probe from the missing pieces.

- Every test now names the rule behind every answer, so a wrong answer can never appear

in my numbers again without the rule that made it standing next to it.

- I also tested ten kinds of creator edit against the engine, found two where it goes completely blind, and fixed both without moving a single number on the real clips.

1. First: which rule actually gave the wrong answers

I ran all 125 testable clips through the pod's rules and recorded which part of the engine produced each answer. This is the full attribution, not a sample:

```
PS C:\Users\nadaa\pdq_for_nada> python attribute_rules.py --engine pdq_engine_ORIGINAL.py

=====
WHICH RULE PRODUCED THE ANSWERS (engine: pdq_engine_ORIGINAL.py)
=====

WRONG answers reproduced (these rules are the problem): 7
SCAN      4
P0         1
P9E        1
P8         1
SCAN      Krrish 3          avg=6.0 frames=27
P0        Smile            avg=26.8 frames=5
SCAN      Captain America: Civil War avg=12.0 frames=4
P9E       Grave Encounters 2 avg=34.0 frames=2
SCAN      The In Between   avg=18.0 frames=4
SCAN      The Northman      avg=6.0 frames=6
P8        The Last Voyage of the Demeter avg=41.4 frames=31

CORRECT answers (these rules must keep working): 52
SCAN      46
P0         3
P3.5      2
P1         1
```

Two things fall out of this picture.

- **The pre-scan is behind 4 of the 7 wrong answers**, at distances 6 to 18. Those are genuinely identical pictures. The rules never ran on them, which is why two weeks of rule changes could never have touched them.
- The other three (P0, P9E, P8) are exactly the rules my earlier changes went after.

2. What the system has that the deciders don't

The engine has three parts that can answer: the pre-scan, the probe (P0), and the rules. The odd thing is how unevenly they are equipped:

check	pre-scan	probe P0	the careful rules (P4, P6a, P7)
sees the whole clip	yes	no, first 4 seconds	yes
needs more than one frame	3	1	5 to 30
hits must agree where in the film they land	no	no	yes
hits from more than one moment of the clip	no	no	some

The engine's own notes say a true match gives the same film-time offset for every matched frame. That is the strongest structural signal in the data, and the two parts that decide almost everything ignore it completely. The pre-scan decides with the probe's confidence and none of the rules' checking.

3. The probe, rebuilt from the missing pieces

pdq_engine_PROBEV2.py. It keeps the pre-scan's whole-clip view (40 sampled frames) and adds everything in that table: all crop families, a mini scorecard per film, the film-time agreement check, the clip-spread check. Its three accept bars each copy a rule the pod already trusts:

bar	fires on	copied from
PV-A	two near-perfect frames	P2.5
PV-B	the old pre-scan bar, plus two hits agreeing on film time	pre-scan + P4
PV-C	five medium frames, agreeing, from two moments of the clip	P6a

The old one-frame P0 is retired in this version; the probe replaces it.

4. The measurement

engine	wrong answers	correct answers
the pod's rules	7	52
the new probe alone, nothing else running	1	41
the new probe, rules as the fallback	1	48

```
PS C:\Users\nadaa\pdq_for_nada> python attribute_rules.py --engine pdq_engine_PROBEV2.py

=====
WHICH RULE PRODUCED THE ANSWERS (engine: pdq_engine_PROBEV2.py)
=====

WRONG answers reproduced (these rules are the problem): 1
  PV-A      1
  PV-A      Krrish 3                      avg=6.0 frames=16

CORRECT answers (these rules must keep working): 48
  PV-A      22
  PV-B      17
  P1         4
  P3.5       3
  PV-C       2
```

- Six of the seven wrong answers are gone. The survivor is Krrish 3 at distance 6: the pictures really are in the index, so nothing picture-based can refuse it. It belongs with audio and SSCD, with its scorecard.

- The four lost correct answers are all montage clips: the right film was the probe's top hint every time, but the hits scatter across film time because the clip cuts between scenes, and then the streaming pass gives up before reaching the matching stretch. I can recover them, but the same loosening lets The Northman and Civil War back in, because on picture evidence those wrong answers look exactly like these right ones. On "a no answer beats a wrong one" I left them out. That is a policy call, and I would like it confirmed rather than mine.

- One oddity from the full tables: the clip the pod answered as The Misfits is answered

as F1: The Movie by both the pod's rules and mine, at distance 8. Worth watching that clip; the live answer may have come from a different detective than we think.

5. The edits test

Ten edits a reposter actually makes, each applied to 8 clips the engine identifies correctly unedited. So when an edited copy fails, the edit is the only reason. First finding: **no edit ever produced a wrong film, on any engine version. Edits only make the engine go silent.**

Filters, black and white, recompression and speed changes barely scratch it. Rotation and zoom degrade it. Two edits blinded it completely: mirroring (0 of 8) and letterboxing (1 of 8). Both turned out to be missing-tech, not thresholds:

- **Mirror:** nothing in the engine ever hashes a flipped frame. The probe now hashes the mirror of every view it samples. Mirrored clips: 0 of 8 to 8 of 8, matching at distance 2.

- **Letterbox:** the pod has an "active region crop" that finds the picture inside the black bars, and it is one of the pieces the laptop port left out. I built a simple version into the probe. Letterboxed clips: 1 of 8 to 6 of 8. The two still missed have bars that are not quite black after recompression.



```
PS C:\Users\nadaa\pdq_for_nada> python test_edits.py --engine pdq_engine_PROBEV2.py --probe-only
```

```
=====
HOW EACH EDIT FARES (engine: pdq_engine_PROBEV2.py, probe only)
=====
edit      right film wrong film no answer rules used
caption    5         0         3  PV-Ax3,PV-Bx2
filter     8         0         0  PV-Ax4,PV-Bx3,PV-Cx1
gray       8         0         0  PV-Ax4,PV-Bx4
letterbox  6         0         2  PV-Ax4,PV-Bx2
lowq       7         0         1  PV-Ax4,PV-Bx3
mirror     8         0         0  PV-Ax4,PV-Bx4
rot2       1         0         7  PV-Cx1
speed080   8         0         0  PV-Ax4,PV-Bx4
speed125   6         0         1  PV-Ax4,PV-Bx2
zoom120    3         0         4  PV-Bx2,PV-Cx1
```

```

PS C:\Users\nadaa\pdq_for_nada> python 3_identify_clip.py <mirrored / letterboxed Magadheera clips>
--- before the fix (test run on the same two clips) ---
mirror      Magadheera          no answer    PROBE_NO_MATCH
letterbox    Magadheera          no answer    PROBE_NO_MATCH

--- after the fix ---
[PROBE-V2] PV-A HIT in 11.8s: vidx=49 best=2 frames_under_16=6 -> early accept
mirror      Magadheera          -> 226907:Magadheera (tmdb=23790)  PV-A: best=2
[PROBE-V2] PV-A HIT in 17.1s: vidx=49 best=8 frames_under_16=5 -> early accept
letterbox    Magadheera          -> 226907:Magadheera (tmdb=23790)  PV-A: best=8

```

The safety gate: I re-ran all 125 real clips with the new views switched on. **Not one answer changed.** The extra views only ever add, never subtract.

The honest cost: the extra views make the probe about four times slower per clip. On my laptop that is irrelevant. If this ever goes near the pod, the sensible shape is plain views first and the mirror and bar-trim views only when the plain views found nothing, so ordinary clips cost what they cost today.

6. Housekeeping

- Everything is in a private GitHub repo now, pdq-testing, with the full history and

all the result CSVs. Happy to add you to it.

- Pod access works end to end from my machine. Verified read-only, nothing touched:

```

PS C:\Users\nadaa\pdq_for_nada> ssh (read-only hello, via the pod access doc's pattern)
ZQ hello 539a9e496adc
ZQ ready {"error":null,"ready":true,"state":"ready","state_age_s":18310.3,"uptime_s":18621.8}

```

- While setting that up, the access doc with the key in it ended up pasted into a chat.

If you would rather rotate the key, now is a clean moment.

7. What I would like from the meeting

1. The policy call on the four montage clips: keep the strict trade, or recover them and accept Northman and Civil War back. I recommend keeping the strict trade.

2. A look at the Misfits clip that both engines call F1: The Movie.

3. A direction: is the rebuilt probe something you want taken towards the pod, as a replacement for the pre-scan and P0? If yes, I would start with the plain-views-first version and measure it on the pod's index size.

4. Still open from last week: retention on clip_query_hashes for voted clips. The clip rot has not stopped being true.

Everything here ran on my laptop against the 134-film index with 16.7 million fingerprints. The full write-up with every table is PROBE_VS_SYSTEM.md; the three commands that reproduce all of it are attribute_rules.py, make_edited_clips.py and test_edits.py. The screenshots are the real printouts of those

runs.